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Printed Page: 1 of 1

Subject Code: BMC204

Roll No:

MCA

(SEM. II) THEORY EXAMINATION 2025-26

DATABASE MANAGEMENT SYSTEMS

TIME: 3 HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****2x7 = 14**

QNo.	Question	CO	Level
(a)	Define database system and its components.	1	K2
(b)	What is data model?	1	K2
(c)	Define relational model.	2	K2
(d)	What is SQL?	2	K2
(e)	Define functional dependency.	3	K2
(f)	What is transaction?	4	K2
(g)	Define locking.	5	K2

SECTION B**2. Attempt any three of the following:****7 x 3 = 21**

(a)	Explain DBMS architecture and data independence.	1	K3
(b)	Explain relational algebra operations with examples.	2	K3
(c)	Explain normalization up to 3NF with examples.	3	K3
(d)	Explain serializability and types of schedules.	4	K3
(e)	Explain concurrency control using locking techniques.	5	K3

SECTION C**3. Attempt any one part of the following:****7 x 1 = 7**

(a)	Explain database system vs file system with advantages of DBMS.	1	K4
(b)	Explain ER model concepts including keys and mapping constraints.	1	K4

4. Attempt any one part of the following:**7 x 1 = 7**

(a)	Explain SQL queries, joins, and aggregate functions.	2	K4
(b)	Explain integrity constraints and types of keys in relational model.	2	K4

5. Attempt any one part of the following:**7 x 1 = 7**

(a)	Explain functional dependencies and lossless join decomposition.	3	K4
(b)	Explain normalization using BCNF and multivalued dependencies.	3	K4

6. Attempt any one part of the following:**7 x 1 = 7**

(a)	Explain log-based recovery and checkpoints.	4	K4
(b)	Explain distributed database system and concurrency control.	4	K4

7. Attempt any one part of the following:**7 x 1 = 7**

(a)	Explain timestamp-based protocol and validation-based protocol.	5	K4
(b)	Explain multiversion concurrency control and Oracle case study.	5	K4